

Ayush Yadav

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Skills

Languages: C/C++, Python, Go, Bash, TypeScript, SQL

Systems & Low-Level: Linux, Systems Programming, Performance Profiling, Memory/Heap Profiling, Concurrency, gdb, Valgrind, AddressSanitizer

EDA & Hardware/Software: Xcelium, SystemVerilog, RTL Simulation, RTL Optimization, Simulation Debugging, Waveform Analysis, Pre-Silicon Validation

Infrastructure: Docker, CI/CD, Jenkins, Git, REST, gRPC, Kafka

AI Engineering: LLMs, AI Agents, Multi-Agent Systems, RAG, MCP, OpenAI Agents SDK

Experience (4.5+ Years)

Cadence Design Systems, Noida

July 2022 - Present

R&D Software Engineer II

- Optimized **Xcelium Logic Simulator** using low-level C++, scalable profiling, and runtime analysis, improving simulation throughput by **18–19%** on large customer workloads.
- Designed and implemented reusable graph-based RTL transformations in the **Xform Engine** operating on large hardware designs and improving runtime performance by **13–14%** for designs from **Apple, Google, Samsung, and NVIDIA**.
- Built and presented a **C++ performance-analysis library** at Cadence India Conference, enabling Top-N profiling and structured diagnostics that reduced RCA time by **~40%**.
- Built **MAESTRO**, a portable multi-agent platform with a conductor orchestrating 10 AI agents across a 20-phase human-gated workflow; measured across **22 Jiras**, cutting ticket-resolution time by **45%** and RCA time by **55%**.
- Built a **~16K-line Xcelium regression system** using rule-based classification, TF-IDF clustering, historical learning, and JEDAI LLM review to automate triage, regolds, waivers, and verification; processed **2,120 failures across 79 runs**, with **73% of failures** were processed in runs by other engineers.
- Built **cross-stream Perforce replay tooling** that derives source streams and selectively syncs changelist-touched paths, automating multi-CL backports across Xcelium release branches.

Projects

- Chipstack Xcelium AI Agents – Performance & Memory Optimization**
 - Built multi-agent Xcelium optimization workflows using **Python, OpenAI Agents SDK, and RAG** to profile and automate performance/memory fixes; achieved **31.3% simulation speedup** on a validated benchmark.
 - Re-architected Xcelium agents with a **deterministic-first Python pipeline**, reducing end-to-end runtime **148.3s to 76.8s (1.93x)** and analysis **71.1s to 18.0s (3.9x)** on a benchmark design.
 - Built a shared **graph-based agent framework** with typed state, checkpoint/resume, structured outputs, fallback recovery, and **YAML-driven LLM-as-judge regression testing**.
- AlgoLens - Interactive DSA Platform**
 - Built an **algorithm visualizer** (60+ algos) with real-time step-through, deterministic replay, & **FFmpeg-WASM export**.
 - Built GenAI explanations, **complexity analysis**, and **exact, fuzzy, phonetic, and semantic search**.
 - Improved quality with **WCAG 2.1 AA** accessibility, **Sentry**, Web Vitals, Playwright, Storybook, Percy, and CI gates.

Achievements

- MAESTRO recognized as proprietary IP and selected for **Trade Secret protection**.
- Corporate VP & MD Recognition** for Xcelium performance gains contributing to a **multi-million-dollar Samsung deal**.
- Competitive Programming:** CodeChef 6-star (2353, AIR 29), Codeforces Master (2264, AIR 13), LeetCode Knight (2124, Top 1%).
- Awarded **\$10,000** to expand a decentralized gaming application on Tezos.

Education

Netaji Subhas University of Technology (NSUT) East Campus

Aug 2018 - May 2022

Bachelor of Technology (B.Tech) in Computer Science and Engineering

CGPA: 8.4 / 10